1 Introduction to Signalling Power Supplies

1.1 Need for Power Supplies

British railways rely heavily on the use of electricity to operate their signalling equipment. In many cases, a direct current supply is required and it is obtained from either of these two sources:

- a battery. An example is shown in figure 1
- a transformer rectifier. An example is shown in figure 2

When an alternating current supply is required it is supplied through a transformer.

1.2 Batteries

A battery is a self-contained power source which produces electricity by a chemical reaction within a cell. By connecting an appropriate number of cells together, you can produce a battery of the required output voltage. There are two types of battery, namely:

- primary batteries
- secondary batteries

You will learn from this training module how both types of battery work and the theory behind their maintenance and testing.

1.3 Transformers

A transformer is used to supply an alternating voltage to electrical equipment and circuits. You will learn how a transformer works.

1.4 Transformer Rectifiers

A transformer rectifier produces a direct current source of power from an AC input. You will learn how this is achieved.

1.5 Safety

Power supplies, like any other electrical equipment, are perfectly safe if you observe the correct safety precautions. This module will remind you why safety precautions are necessary when handling power supplies.

2 The Electrical Cell

2.1 Construction of the Electrical Cell

An **electrical cell** produces a voltage by chemical action. It contains two **plates** immersed in an **electrolyte.** The two plates are called **electrodes** and each electrode is made of a different material. Figure 3 shows the construction of a simple cell.

A chemical action between the electrolyte and electrodes results in one electrode receiving a **negative** (-) electrical charge and the other a **positive** (+) charge. It is the difference between the positive and negative charges which produces the cell's output voltage.

The **output voltage** will depend on the type of cell but a typical voltage for a single cell is between 1.2 and 1.4 volts. The **output current** will depend on the load connected to the cell.

2.2 Connecting Cells Together

Cells are connected together to supply a required voltage or current. You can connect them in one of two ways, namely:

- in series to supply a voltage which is higher than that of a single cell
- in parallel to supply a current which is higher than that available from a single cell

Construction of a Simple Cell

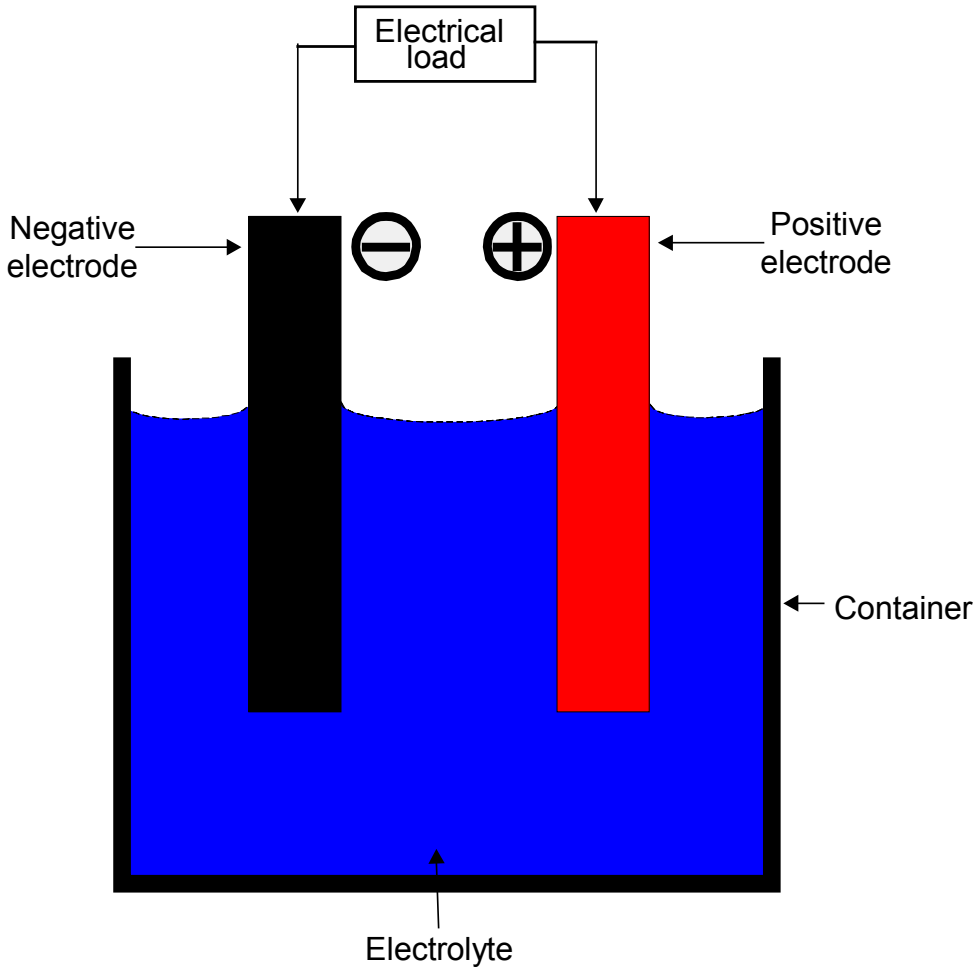


Figure 3

Series Connection

Series connection means connecting directly from one cell to the next, then the next and so on. The connection is made from the positive (+) terminal of one cell to the negative (-) terminal of the next cell.

Figure 4 is a circuit diagram of three cells connected in series.

Assume that each cell provides 1.4 volts, then by connecting three cells in series you can obtain a voltage of 1.4 + 1.4 + 1.4 = 4.2 volts. The current through each cell is the same.

So, by connecting cells in series you can increase the voltage output.

Parallel Connection

Parallel connection means connecting the positive terminal of one cell to the positive terminal of another. Similarly, the negative terminals are connected together.

Figure 5 is a circuit diagram of three cells connected in parallel.

Assume that each cell can provide 1.4 volts and 50 milliamps. By connecting three cells in parallel you can obtain an output current of 50 + 50 + 50 = 150 milliamps. The voltage output is unchanged at 1.4 volts.

So, the advantage of connecting cells in parallel is that more current can be supplied to an electrical circuit, without drawing too much current from an individual cell.

Series and Parallel Connection of Cells

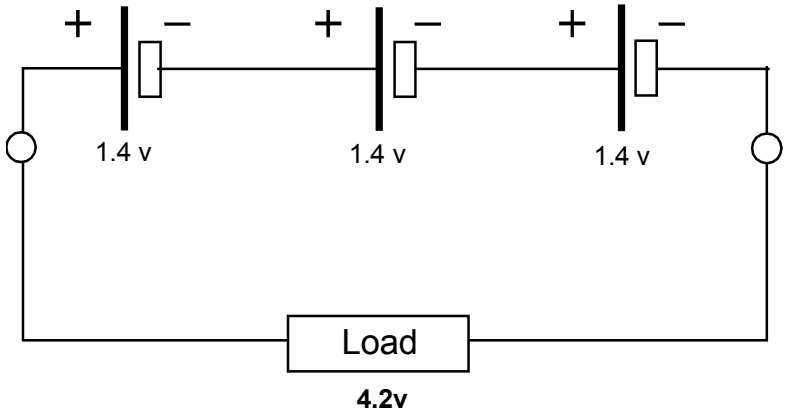


Figure 4 - Series Connection of Cells

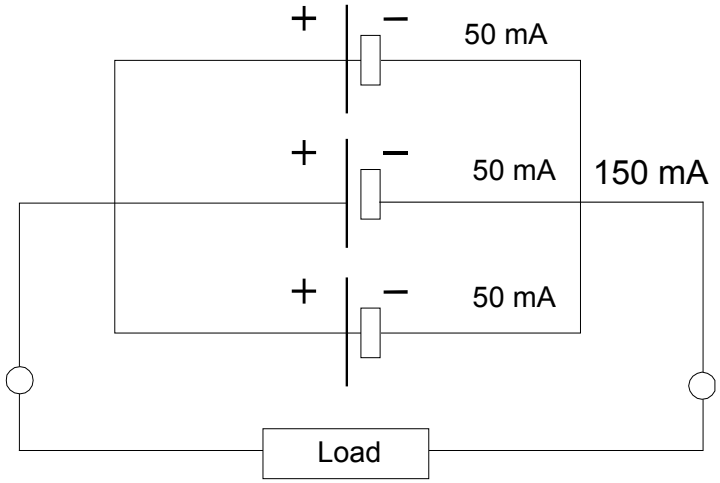


Figure 5 - Parallel Connection of Cells

An example of the use of cells connected in parallel is the track circuit. Parallel cells are used because together they last longer. Other advantages are

- one cell can be removed and replaced without interrupting the operation of the track circuit
- the track circuit will still function if one cell fails because power will continue to be provided by the other cell

2.3 Internal Resistance of a Cell

Whenever a cell provides an electric current to flow through a circuit, energy is wasted in overcoming the resistance of the circuit. Every part of the circuit has some resistance and this includes the cell itself. The resistance inside a cell is called the internal resistance.

You know that a current flowing through a resistance will cause a drop in voltage. This is also true of a cell's internal resistance. Therefore, not all the voltage produced by a cell is available to the circuit being supplied. This is shown in figure 6.

The internal resistance reduces the voltage available at the cell's terminals.

Voltage lost to internal Resistance of a cell

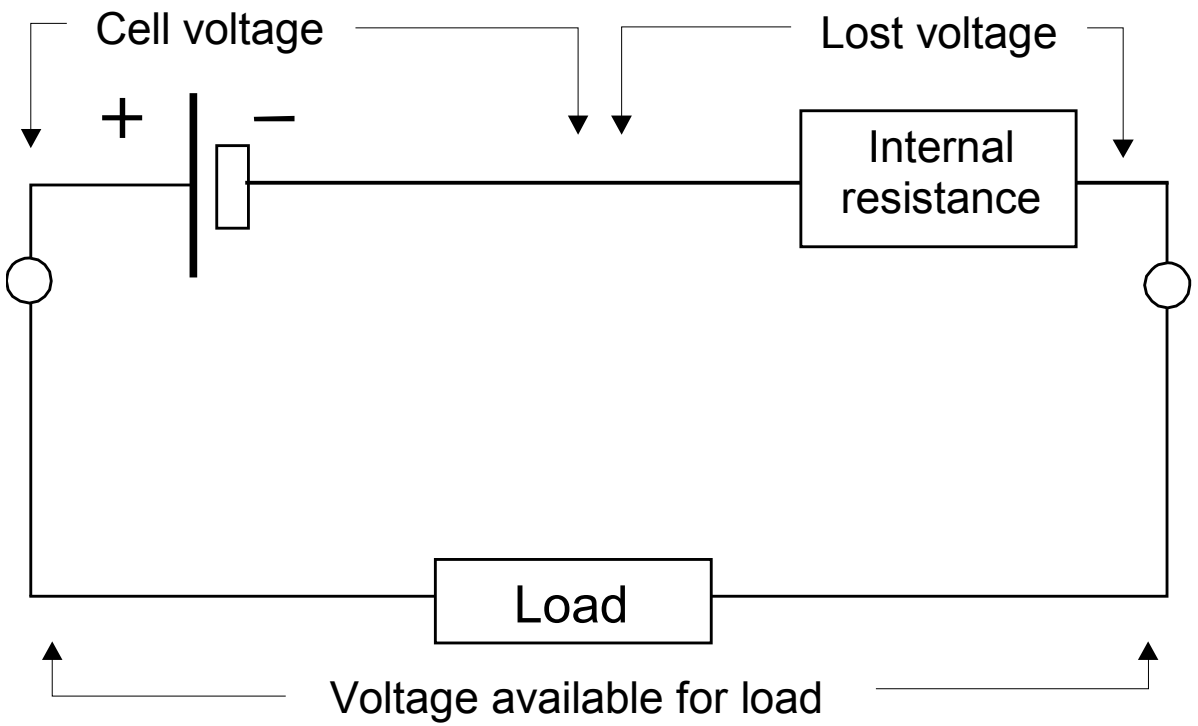


Figure 6

2.4 Self-Assessment Questions

1. Cells are connected in series to provide a higher voltage. Is this true?
2. Cells are connected in parallel to provide a higher current. Is this true?
3. Assume that you have a supply of cells each of which can supply 1.4 volts. How many cells would you need to connect in series to obtain a supply of 7 volts?
4. If one cell can provide an output voltage of 1.4 volts with a current of 100 milliamps, how many cells would you need to supply 1.4 volts and 500 milliamps? How would you connect them together?
5. Sketch the symbol for an electrical cell.
6. What effect does the internal resistance of a cell have on the cell's terminal voltage?
7. What is the difference between a battery and a cell?

If you are not totally confident that you have correctly answered any of these questions, re-read the relevant parts of this section to check your answer(s) before going onto the next section.

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3 Primary Cells

3.1 Construction of a Primary Cell

The construction of a typical **primary cell** is shown in figure 7. The type of cell shown is known as a **Leclanche** cell.

A potential problem in a primary cell is the forming of hydrogen gas bubbles around the positive electrode. This is called **polarisation**; it increases the **internal resistance** and so reduces the terminal voltage output from the cell. To prevent polarisation, a **depolariser** compound is placed around the positive electrode to absorb the hydrogen.

All the cell components are encased in an **outer case** with only a small part of each electrode protruding for connection to the cell **terminals**.

A primary cell cannot be recharged to reverse the chemical action . After it becomes 'flat', the cell must be discarded and replaced with a new one. You must comply with the appropriate instructions for the disposal of primary cells .

Construction of a typical Primary Cell

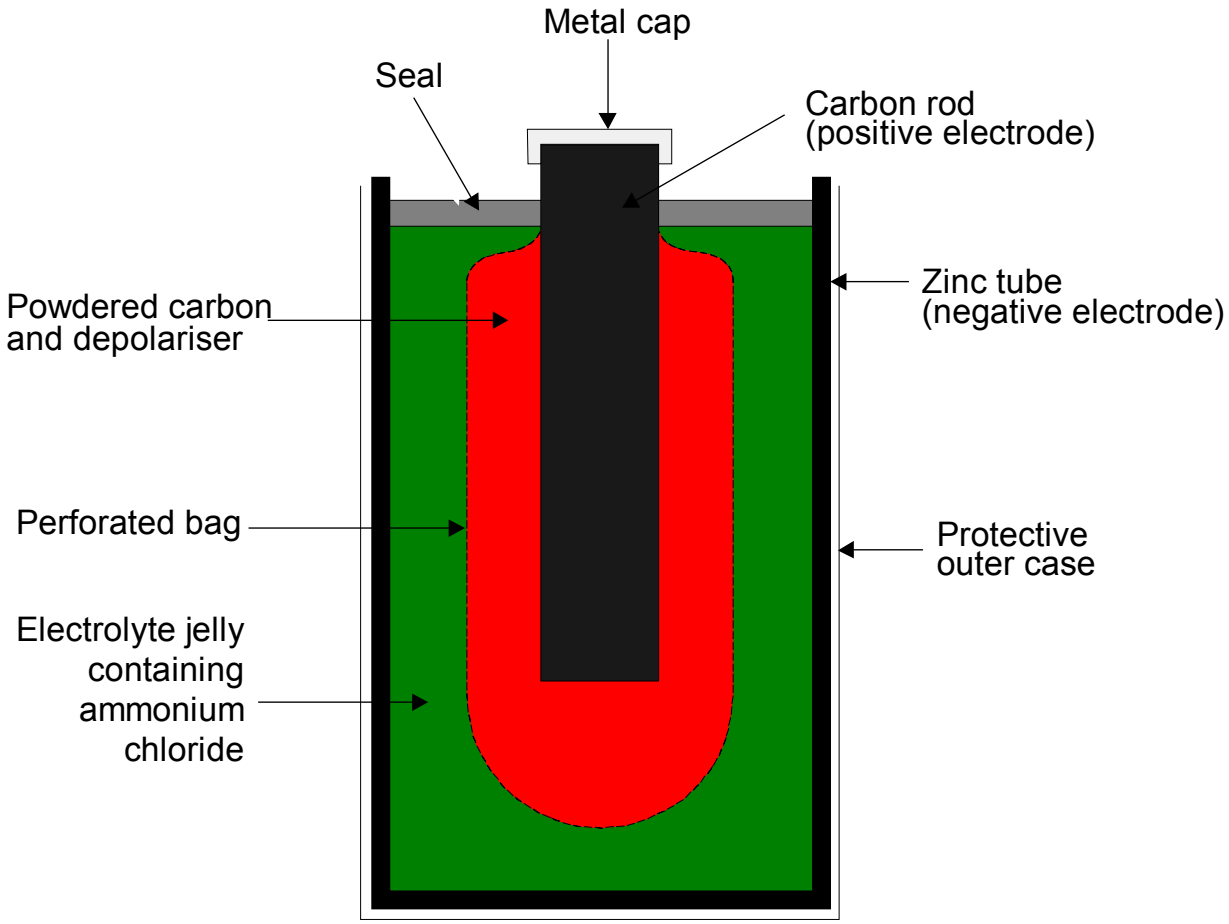


Figure 7

3.2 Use of Primary Cells

Primary cells are used where there is no mains electricity supply available.

3.3 Types of Primary Cell

A primary cell may be dry or wet. The differences between the two are:

- A dry cell has an electrolyte in the form of a thick paste that does not spill and so allows the cell to be used in any position.
- A wet cell has a liquid electrolyte that will spill if the cell is turned upside down. It is often manufactured and transported dry and then water added before it is used.

You will find both types of cell used in the Railways signalling system, namely:

- Leclanche Dry Cell. It has a carbon positive electrode and zinc negative electrode. The cell is issued ready for use with a working voltage of about 1.4 volts.
- Inert Caustic Soda Cell. It too has carbon and zinc electrodes with caustic soda electrolyte. It is a wet cell and you must fill it with water before you can use it. The working voltage is about 1.2 volts. Caustic soda is dangerous and must be handled with care.

A Leclanche Dry Cell and an Inert Caustic Soda Cell are shown in figure 8.

Leclanche Dry Cell (left) and Inert Caustic Soda Cell (right)



Figure 8

Use of Primary Cells

PRIMARY CELLS IN THE WORKPLACE			
TYPE	**DRY**	**WET**	**APPLICATION**
DS1	✓		Track circuit repeating

Figure 9

3.4 Assignment 1

You should be able to find examples of primary cells used in your workplace. Speak to your mentor and work mates, and use your own experience, to identify the types and uses of primary cells in the signalling system. Write your findings in the table in figure 9.

3.5 Testing Primary Cells

It is important that the condition of a cell is determined to assess if it is fit for service. You can assess a primary cell in two ways, both of which you will experience in your practical training. The two ways are:

- By a physical inspection of the cell.
- By a voltage check.

Physical Inspection

The inspection which you carry out for a primary cell will include the following checks.

Remember, before inspecting any cell you must refer to the appropriate safety procedures.

- Adequate **ventilation**. It is very important that the housing containing the primary cells provides a continuous supply of fresh air to flow across the top of the cells. Without a good supply of fresh air, the inert caustic soda cell and some types of Leclanche cell will not operate satisfactorily.

- **Corrosion** of the cells. The chemical action of the battery, and possible chemical leakage, will corrode the cells and their housing.
- **Leaking cells**. Chemicals which leak from cells can be dangerous and cause injury to you and damage to equipment.
- **Distortion of cells**. A cell which is distorted is potentially dangerous because it may explode and injure you. If the outer casing splits it will leak chemicals.
- **Cleanliness** and **security** of terminals. Dirty or loose terminals lower the voltage output of a cell.

Voltage Check

Remember, before checking the voltage of a cell you must refer to the appropriate safety procedures.

The method of checking the terminal voltage of a primary cell will depend on the load which the cell is supplying. There are two types of load that you will encounter, namely:

- **Constant load**, that is a load which continually draws current from the cell.
- **Intermittent load**, that is a load which occasionally draws current from a cell.

Testing a Constant Load

The voltage of a cell which has a constant load is measured with the cell connected to the load. The load will be drawing current and some voltage will be lost because of the internal resistance of the cell. Therefore, the voltage which you measure at the cell's terminals will be the true voltage that is delivered to the load.

The value of the measured voltage at the cell's terminals will depend on the condition of the cell. Typical terminal voltage values are:

- A **new dry cell** will have a voltage of about **1.4 volts**.
- A **discharged dry cell** will have a voltage of about **0.9 volts**.
- A **new wet cell** will have a voltage of about **1.2 volts**.
- A **discharged wet cell** will have a voltage of about **1.0 volts**.

Notice the difference between the dry and wet primary cells. The output voltage of a wet cell will drop quite quickly from its new value. The dry cell has a much slower fall-off characteristic.

Testing an Intermittent Load

The problems with an intermittent load are that if the cell voltage is measured when there is no load connected to the cell, then:

- **No current** will be drawn from the cell.
- There will be **no voltage** loss across the cell's internal resistance.
- The **measured voltage** will not be a true indication of the condition of the cell.

Therefore, the voltage of a cell which has an intermittent load is measured with a 1 ohm shunt resistor connected across the cell with the load still connected. The shunt resistor allows the voltage to be measured in the same way as for a constant load.

3.6 Assignment 2

You should be able to find examples of both constant and intermittent loads in your workplace. Speak to your mentor and work mates, and use your own experience to identify examples of the two types of load. Write your findings in the tables at figure 10.

Having identified the primary cells you use in your workplace, use the table in figure 11 to write down the charged and discharged voltages for each type of cell. Ask your mentor to help you find this information.

Constant and Intermittent Cell Loads in the Workplace

EXAMPLES OF **CONSTANT** CELL LOADS IN YOUR WORKPLACE
1. 2. 3. 4.

EXAMPLES OF **INTERMITTENT** CELL LOADS IN YOUR WORKPLACE
1. 2. 3. 4.

Figure 10

Cell voltages

TYPE OF CELL	VOLTAGE CHARGED	VOLTAGE DISCAHRGED

Figure 11

3.7 Self-Assessment Questions

1. Sketch a primary cell and label the following parts.

   Electrolyte

   Positive electrode

   Negative electrode

   Depolariser

2. What are the two types of primary cell?

3. What are the two methods of testing a primary cell?

4. What are the two types of load which may be connected to a primary cell?

5. How is the terminal voltage tested for a cell with a constant load?

6. How is the terminal voltage tested for a cell with an intermittent load?

If you are not totally confident that you have correctly answered any of these questions, re-read the relevant parts of this section to check your answer(s) before going onto the next session.

4 Lead Acid Cells

4.1 Construction of a Lead Acid Cell

A lead acid cell is a secondary cell. This means that the cell has a reversible chemical action and can be recharged when it becomes ‘flat’.

The construction of a typical lead acid cell is shown in figure 12. The cell is encased in a container which can be glass or another rigid acid resisting material.

4.2 The Plates

- The **positive plate** is made of lead peroxide (**chocolate brown**).
- The **negative plate** is made of spongy lead (**silver grey**).

When discharged both plates are coated with lead sulphate (whitish grey).

The plates are interleaved and immersed in an electrolyte. To make the cell less prone to damage by physical shock, the plates are usually supported and kept apart by separators.

Construction of a lead Acid Cell

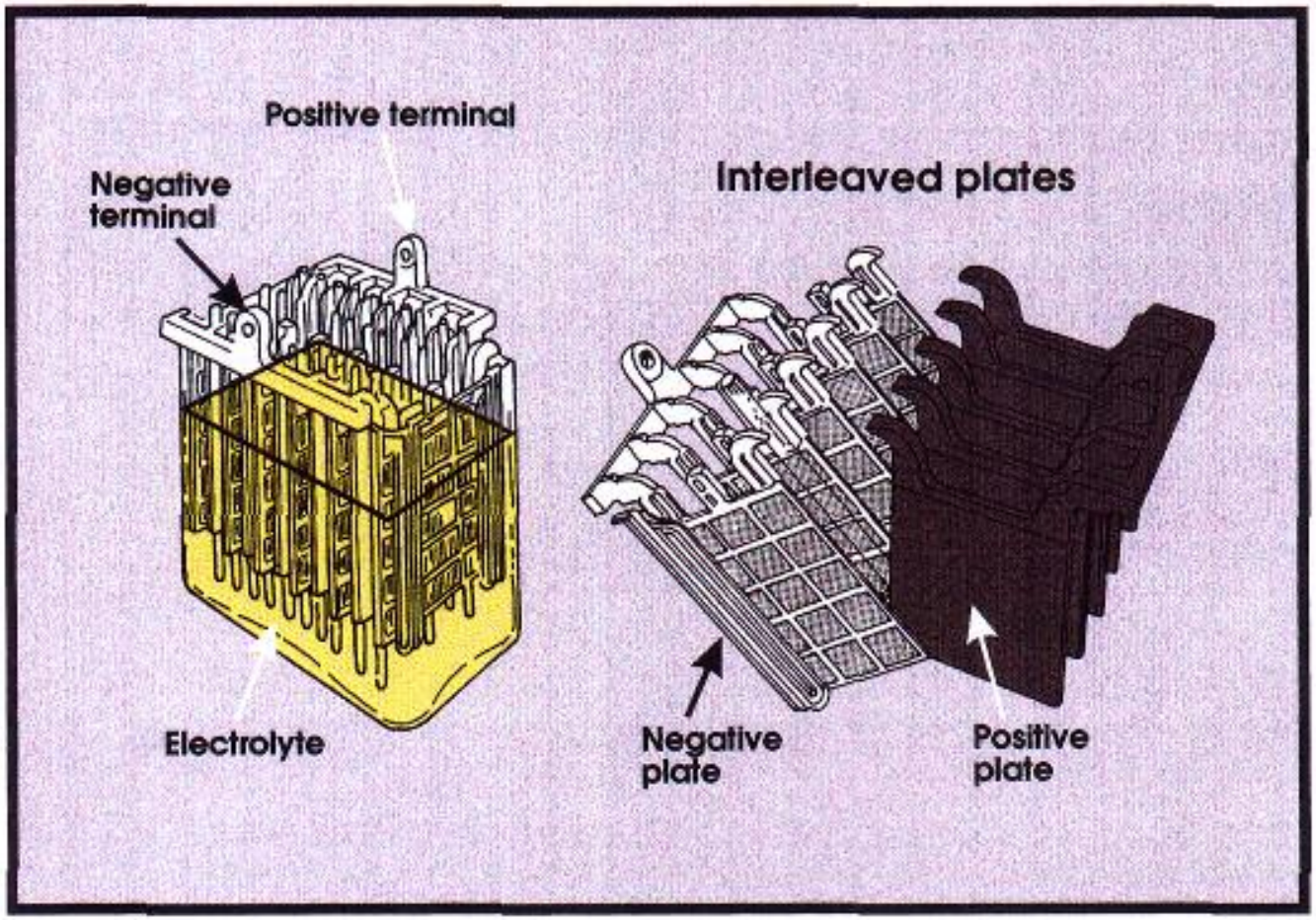


Figure 12

4.3 The Electrolyte

The electrolyte is dilute sulphuric acid. Its strength is checked by measuring its specific gravity. Pure water has a specific gravity of 1.000 and the electrolyte will always be greater than this.

- the **higher** the specific gravity, the stronger the acid in the electrolyte
- the **lower** the specific gravity, the weaker the acid in the electrolyte

The specific gravity of a cell's electrolyte provides you with a very good indication of the electrical charge held by the cell.

Remember, sulphuric acid is dangerous and you must refer to the appropriate safety procedures before handling electrolyte.

4.4 Assignment 4

You should try to find examples find examples of lead acid cells used in your workplace. Speak to your mentor and work mates, and use your own experience, to identify the types and uses of lead acid cells in the signalling system. Write your findings in the table at figure 13 on the next page.

Use of Lead Acid Cells

LEAD ACID CELLS IN THE WORKPLACE	
TYPE	**APPLICATION**

Figure 13

4.5 Testing Lead Acid Cells

It is important that the condition of a cell is determined to assess if it is fit for service. You can assess a lead acid cell in three ways, namely

- by a **physical inspection** of the cell
- by a **voltage check**
- by **measuring the specific gravity** of the electrolyte

4.6 Physical Inspection

Remember, before inspecting a cell you must refer to the appropriate safety procedures.

The inspection which you carry out for a lead acid cell will include the following checks

- **cleanliness** of the cells and their cases
- **cleanliness and security of terminals**. Dirty or loose terminals introduce a high resistance in the circuit and therefore provide a lower voltage to the load. An example of loose terminals causing overheating is shown at figure 14
- **cleanliness and dryness of battery racks and shelves**. A wet and dirty environment is dangerous to you and causes damage by corrosion. Wet shelves can cause earth faults
- the **level of electrolyte** in each cell. Lack of electrolyte lowers the output voltage of the cell

Loose, Overheated Battery Terminals



Figure 14

4.7 Voltage Check

Remember, before checking a cell you must refer to the appropriate safety procedures.

The output voltage of a cell is measured using a voltmeter as shown in figure 15. The method of checking the voltage of a lead acid cell will depend on the load which the cell is supplying. There are two types of load that you will encounter, namely

- constant load, that is a load which continually draws current from the cell
- intermittent load, that is a load which occasionally draws current from a cell

Measuring the Voltage of a Secondary Cell



Figure 15

4.8 Measuring Specific Gravity

Remember, before you measure the specific gravity (SG) of a cell you must refer to the appropriate safety procedures.

You can measure the specific gravity (SG) of the electrolyte using a hydrometer, a picture of which is shown in figure 16. A sample of the electrolyte is sucked into a tube which has a small float inside it. The level of the float indicates on a scale the electrolyte's strength. The float and scale are shown in figure 17.

- A **fully charged** lead acid cell should read about 1.220 on the scale
- A **fully discharged** lead acid cell can read about 1.008 on the scale.

For lead acid cells it is not good practice to allow them to become fully discharged because the lead sulphate which forms on the plates as they discharge can become very thick and resistant to removal by the re-charging process. If left for prolonged periods the lead sulphate can become totally persistent and seriously reduce the electrical capacity of the cell, plus increase its internal resistance. When a cell is trickle charged (see Section 6) it would be evidence of inadequate charging if the SG falls to 1.160 or lower.

Measuring the Specific Gravity of Electrolyte by using a Hydrometer



Figure 16

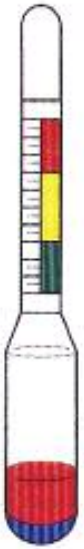


Figure 17.
Hydrometer's Float and Scale

4.9 Self-Assessment Questions

1. Is a lead acid cell a secondary cell?

2. What is the specific gravity of distilled water?

3. What are the three ways of testing a lead acid cell?

4. What is the name of the instrument used to measure specific gravity?

5. What is the approximate specific gravity of a fully charged lead acid cell?

6. What is the approximate specific gravity of a fully discharged lead acid cell?

If you are not totally confident that you have correctly answered any of these questions, re-read the relevant parts of this section to check your answer(s) before going on to the next section.

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5 Alkaline Cells

5.1 Construction of an Alkaline Cell

An alkaline cell is a secondary cell. This means that, in the same way as a lead acid cell, it has a reversible chemical action and can be recharged when it becomes 'flat'. For a given electrical capacity, the alkaline cell is much larger than a lead acid cell and is more expensive. However, its useful life is very much longer than a lead acid cell and it recovers more easily from being fully discharged.

A battery is made up of several cells connected together in series to provide the required voltage. The cells are encased in a steel container. The alkaline cell is more robust than the lead acid cell and therefore more resistant to damage.

A photograph of a typical alkaline battery is shown at figure 18.

The Plates

- The **positive plate** is made of nickel hydrate.
- The **negative plate** is made of iron and cadmium.

Because its plates are of nickel (chemical symbol Ni) and iron (chemical symbol Fe), this type of cell is often referred to as a NiFe cell, pronounced 'knife'. It is also often referred to as a nickel cadmium cell. The plates are interleaved and immersed in an electrolyte.

An Alkaline Battery

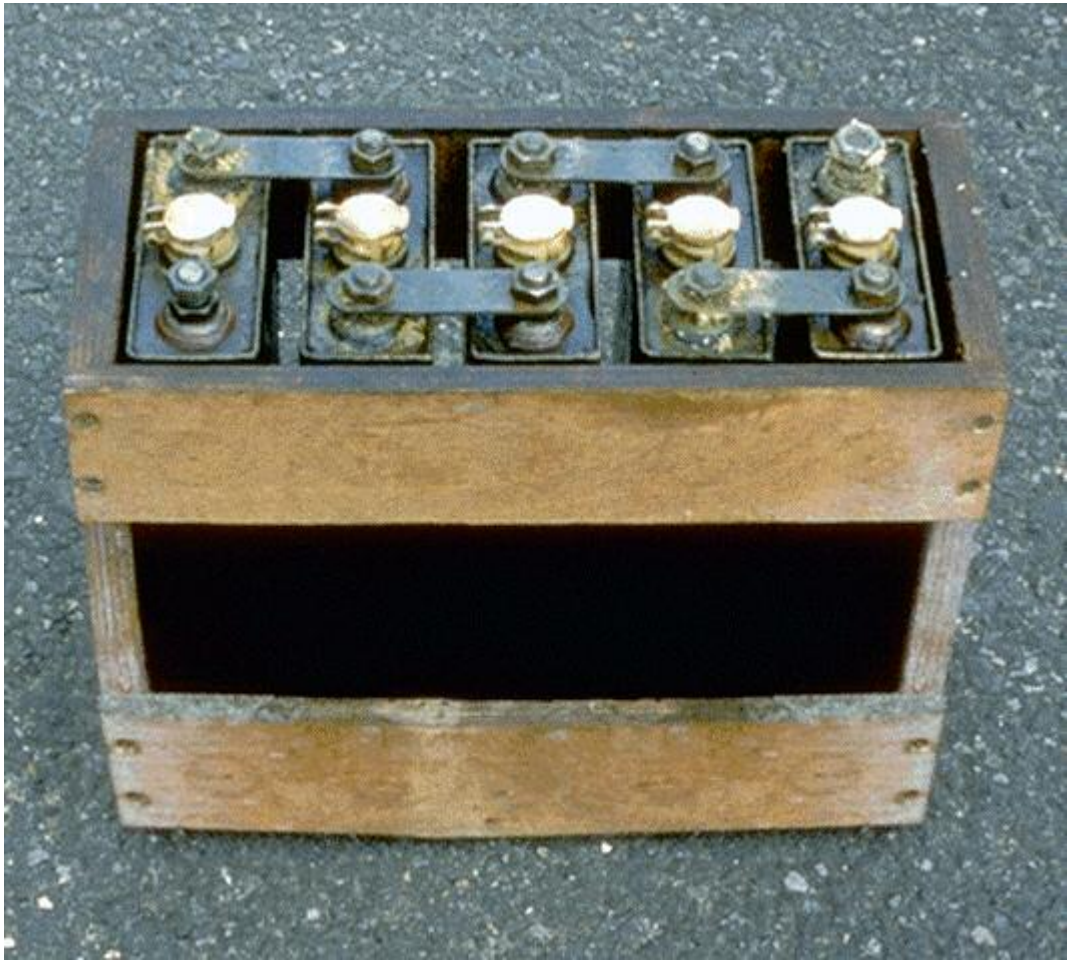


Figure 18

The Electrolyte

The electrolyte of an alkaline cell is dilute **potassium hydroxide**.

Unlike a lead acid cell, the specific gravity of an alkaline cell changes very little as it becomes discharged with use. It remains constant at about 1.190 during use, falling to 1.160 when it is fully discharged.

Because of this small change in specific gravity, measuring the specific gravity of an alkaline cell does not give a good indication as to the state of charge of the cell.

5.2 Testing Alkaline Cells

It is important that the condition of a cell is determined to assess if it is fit for service. Similarly to a lead acid cell, you can assess an alkaline cell in three ways, namely

- by a **physical inspection** of the cell. An example of alkaline cells with corroded terminals is shown in figure 19
- by a **voltage check**
- by measuring the **specific gravity** of the electrolyte

Corroded Terminals on Alkaline Cells



Figure 19

5.3 Assignment 5

You should try to find examples of alkaline cells used in your workplace. Speak to your mentor and work mates, and use your own experience, to identify the types and uses of alkaline cells in the signalling system. Write your findings in the table at figure 25.

5.4 Self-Assessment Questions

1. Is an alkaline cell a secondary cell?

2. What is the specific gravity of a fully charged alkaline cell?

3. What is the specific gravity of a discharged alkaline cell?

4. Why is the specific gravity of an alkaline cell not a good indication of the state of charge of the cell?

If you are not totally confident that you have correctly answered any of these questions, re-read the relevant parts of this section to check your answer(s) before going on to the next section.

Use of Alkaline Cells in the Workplace

ALKALINE CELLS IN THE WORKPLACE	
Type	**Application**

6 Charging a Battery

6.1 Electrical Storage Capacity

The measure of the electrical storage capacity of a cell is the amount of current it can deliver and for how long this current can be maintained before the cell is discharged. Capacity is measured in **ampere/hours** (AH). A 30 AH cell, when fully charged, could deliver:

- 3 amps for 10 hours (3 amps x 10 hours =30 AH), or
- 5 amps for 6 hours (5 amps x 6 hours = 30 AH), or
- any combination of values which equals 30 AH

The capacity of a cell may be quoted as the current which it can supply for 10 hours. This is known as the 10-hour rate. For example, if the 10-hour rate of a cell is quoted as 4 amps, it can supply 4 amps for 10 hours and has a capacity of 40 AH (4 x 10 = 40).

The electrical storage capacity of a cell is reduced:

- if the current drawn from the cell is higher than the 10-hour rate
- as the cell ages

6.2 Self-Assessment Questions

1. If the capacity of a battery is 60 AH, how many amps can it supply for 10 hours?

2. A cell is stated as having a 10-hour rate of 3 amps. What is the capacity of the cell in AH?

If you are not fully confident that you have correctly answered either of these questions, re-read item 6.1 and try again.

6.3 Charging Secondary Cells

Secondary cells or batteries are charged by using a battery charger. On the railway there are a variety of ways in which the charging can be applied, depending on the particular application.

- **Initial Charge**. A secondary cell or battery is usually supplied filled with electrolyte but in a discharged condition. It must be given an initial charge before it can be put into service.
- **Recharge.** During service it is possible that a secondary cell or battery can become so far discharged that it no longer functions. If this happens it has to be taken out of service and recharged. On the railway this is an unusual event and indicates that there is a fault of some kind with the cell or battery or the circuit in which it is being used.
- **Trickle Charge.** Rather than wait for the secondary cell or battery to become fully discharged before recharging it, a battery charger which provides a constant charging current is permanently connected to continuously top up the electrical charge. This method is used when the load is intermittent. The cell or battery delivers the current to the load and the smaller trickle charge current tops up the cell or battery during the periods when the load is off. Provided the trickle charge current is more than the average load current the cell or battery remains fully charged.

- **Float Charge.** When the current demand of the circuit connected to the cell or battery is normally continuous, the battery charger has to deliver the average load current and the cell or battery merely serves to absorb any fluctuations in the current load. In this case the constant current trickle charger is replaced by a float charger which has an almost constant output voltage. In the event that the load is interrupted the float charger behaves like a trickle charger, but when the load is on it delivers not less than the average load current and the cell or battery remains fully charged.

- **Constant Voltage Charger.** When the current demands on the battery are large and extremely variable, such as occur with point machine motors, this special type of float charger is employed. It has an accurately controlled constant output voltage which allows it to provide a very small trickle charge when the battery is fully charged and a large recharge current when the battery is not fully charged.

WARNING

The chemical action which occurs in a cell when it is being charged causes it to give off potentially explosive gases. The gasses can cause violent explosions if adequate precautions are not taken. You must observe all safety precautions before charging a cell or battery.

The charge applied to a cell is normally expressed in one of two ways:

- The charging rate.
- The charging current.

Charging Rate

The charging rate of a cell is the time for which the cell is to be charged and is expressed in hours. The charging current (amps) is calculated using the charging rate (hours) and the cell electrical capacity (AH).

For example, a battery with a capacity of 30 AH which is to be charged at a charging rate of 6 hours will require a charging current of 5 amps (30 AH 6 hours = 5 amps).

A commonly used charging rate is 6 hours.

Charging Current

The charge to a cell may be stated by its charging current. For example, if a battery is to be trickle charged, a charging current of 2 amps may be quoted as the current to be used to continuously charge the battery.

Incorrect Charging

When secondary cells or batteries are being charged it is important that the correct rate of charge is used.

Continuing to charge cells when they are already fully charged causes excessive gassing, serious overheating and heats up the electrolyte, which results in high water consumption and a consequential rapid fall in the electrolyte level.

Using an excessively large charging current causes violent gassing, serious overheating and permanent damage to the plates, which reduces the cells' capacity and shortens their useful life.

An insufficient charge leaves the cells not fully charged, reduces their capacity and shortens their life. On lead acid cells, the build up of lead sulphate on the plates causes serious problems.

6.4 Assignment

In your workplace you may find examples of initial, trickle, float and constant voltage battery charging. Discuss this with your colleagues and your Mentor and identify which types are used in the signalling system and where they are used. Write your findings in the table at figure 21.

Battery Charging in the Workplace

Battery Charging in the Workplace	
Type of Charge	**Application**

Figure 21

6.5 Self-Assessment Questions

1. What unit of measurement is used to state a cell's electrical capacity?

2. This module described three types of charge for a battery. What are they?

3. A cell with a capacity of 40 AH is to be charged at a charging rate of 8 hours. Will the charging current be 3, 5 or 8 amps?

4. How would you recognise that a cell is being charged at an excessive rate?

5. What are the effects of using an insufficient charging rate?

If you are not totally confident that you have correctly answered any of these questions, re-read the relevant parts of this section to check your answer(s) before going on to the next section.

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7 Transformers

7.1 The Transformer

Principles of a Transformer

The transformer works on two principles. They are:

Passing a current through an electrical conductor produces a magnetic field around the conductor. An alternating current will produce a correspondingly alternating magnetic field.

An electrical conductor which is in an alternating magnetic field will have an alternating voltage induced into it.

Construction of a Transformer

The basic construction of a transformer is shown in figure 22. The AC supply voltage produces an alternating current in the primary coil, which produces an alternating magnetic field in the laminated iron core. The alternating magnetic field also passes through the secondary coil and induces an alternating voltage in it to produce an alternating current in the electrical load. Electrical energy is transferred from the primary coil to the secondary coil without any direct electrical connection between them.

Construction of a Transformer

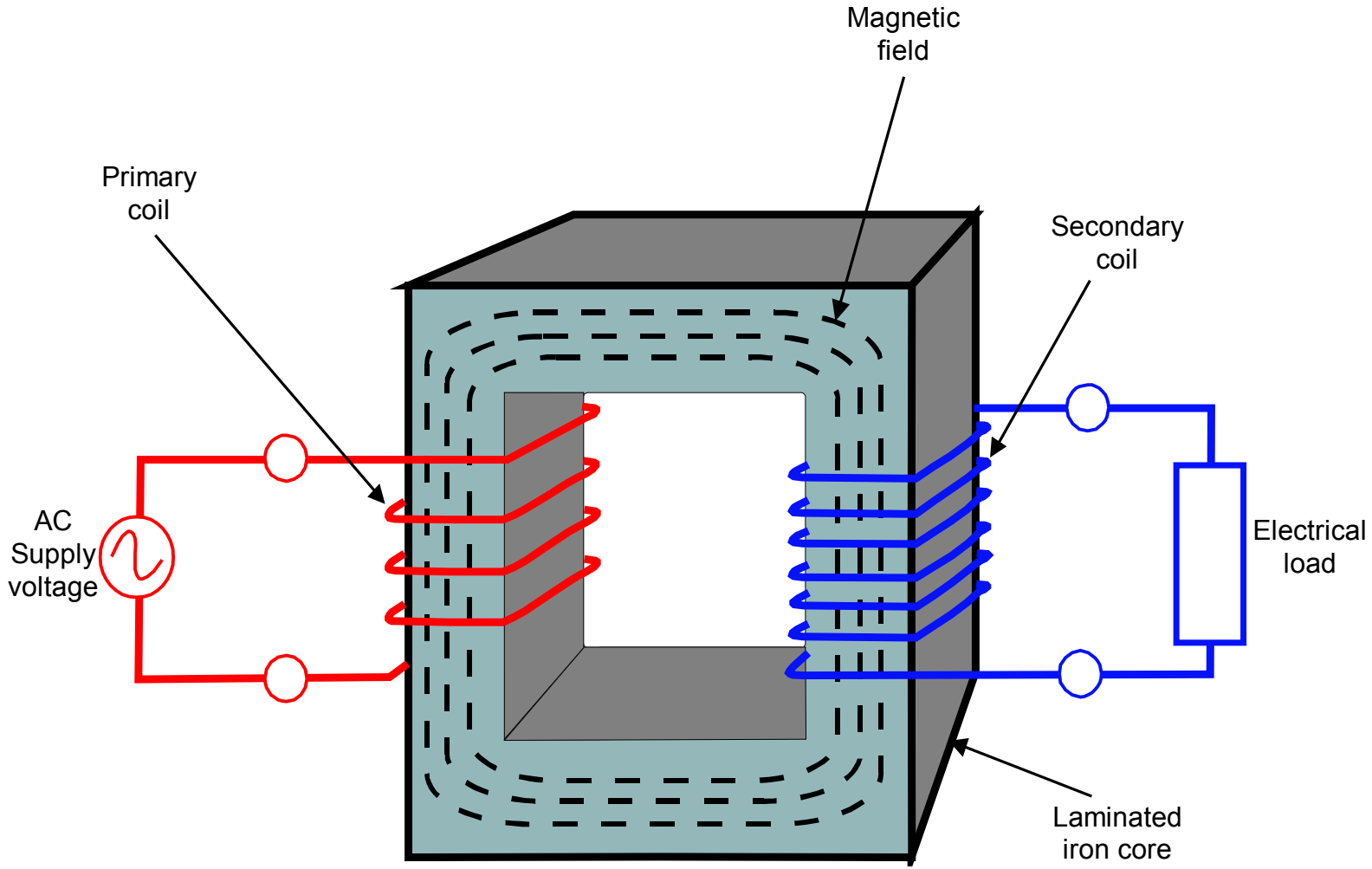


Figure 22

Voltage in the Transformer

The transformer is used to change AC voltage values in one circuit to a different value in another circuit. The change in value will depend on the difference between the number of turns of wire in the primary coil and the number of turns of wire in the secondary coil. It is found that:

If there are more turns of wire in the secondary coil than there are in the primary coil, then the voltage across the secondary coil is greater than that across the primary coil.

If there are less turns of wire in the secondary coil than there are in the primary coil, then the voltage across the secondary coil is less than that across the primary coil.

7.2 Voltage Control

The change in AC voltage from primary coil to secondary is directly proportional to the ratio of the number of turns in the two coils. The following examples will illustrate this:

- If a transformer has 1100 turns in the primary coil and 120 turns in the secondary coil, then if 110 volts is applied to the primary there will be 12 volts in the secondary coil.

- If a transformer has 650 x n turns in the primary coil and 110 x n turns in the secondary coil, then if 650 volts is applied to the primary there will be 110 volts in the secondary coil.

These two examples are illustrated in figure 23.

Controlling Voltage Output

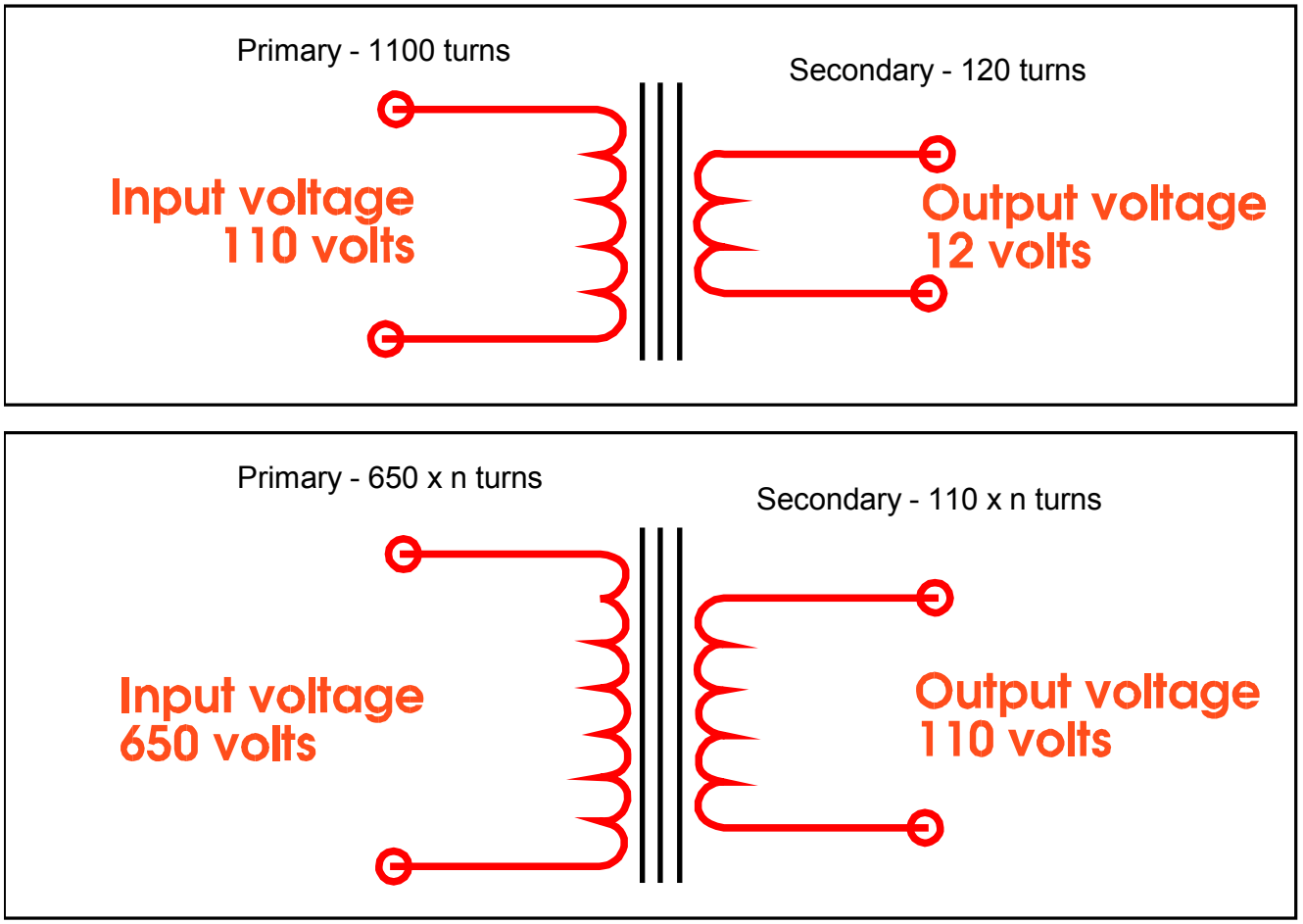


Figure 23

The ratio of the number of turns in the primary coil to the number of turns in the secondary coil of a transformer is called the turns ratio. By changing the turns ratio, the secondary voltage can be controlled.

The illustration at figure 24 shows **voltage control** in the secondary coil of a transformer. The control is used to change the turns ratio by changing the number of turns in the secondary coil and thus control the output voltage.

One method of voltage control is to use a **switch** to change the number of turns; an example is a manually controlled battery charger. Another method uses a **tapped transformer** to provide a choice of voltage outputs to which the load can be connected; this method is used to supply the required alternating voltage to electrically lit signals.

7.3 Isolating Transformers

All modern signalling circuits are arranged to be **earth free** because

- the earth is a good conductor of electricity
- if two conductors were to come into contact with the earth, signalling equipment could be falsely operated

It is normal practice for one wire of the public electricity supply to be earthed and so an **isolating transformer** is used to electrically isolate non-earthed signalling equipment from the earthed electricity supply.

An illustration of a transformer with voltage adjustment tappings.

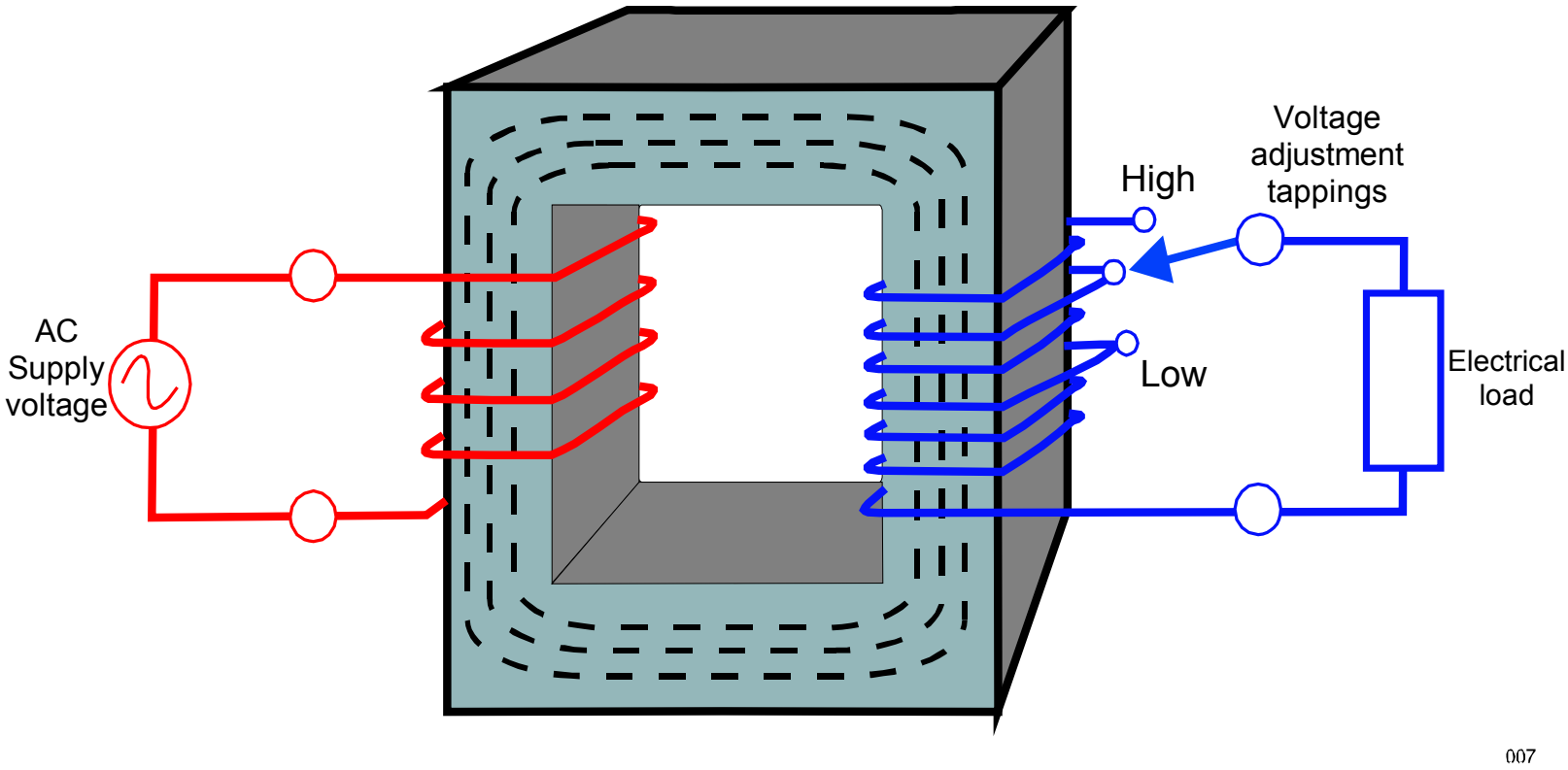


Figure 24

7.4 Transformer Design

The power output from the secondary coil is almost equal to the power input to the primary coil. The difference is the small power loss in the iron core.

The power which a transformer can safely handle is mainly determined by its physical size. A high power transformer requires thicker wire for the coils and a larger core, therefore increasing the overall size and weight.

A transformer's coils normally have more than one turn of wire for every volt which is input or output. One of the main factors which determines the number of turns of wire per volt is the frequency of the alternating voltage for which the transformer is designed. A typical transformer designed for use on the 50Hz mains supply would have 8 turns per volt.

A transformer with voltage adjustment tappings is shown in figure 25.

Transformer with Voltage Adjustment Tappings

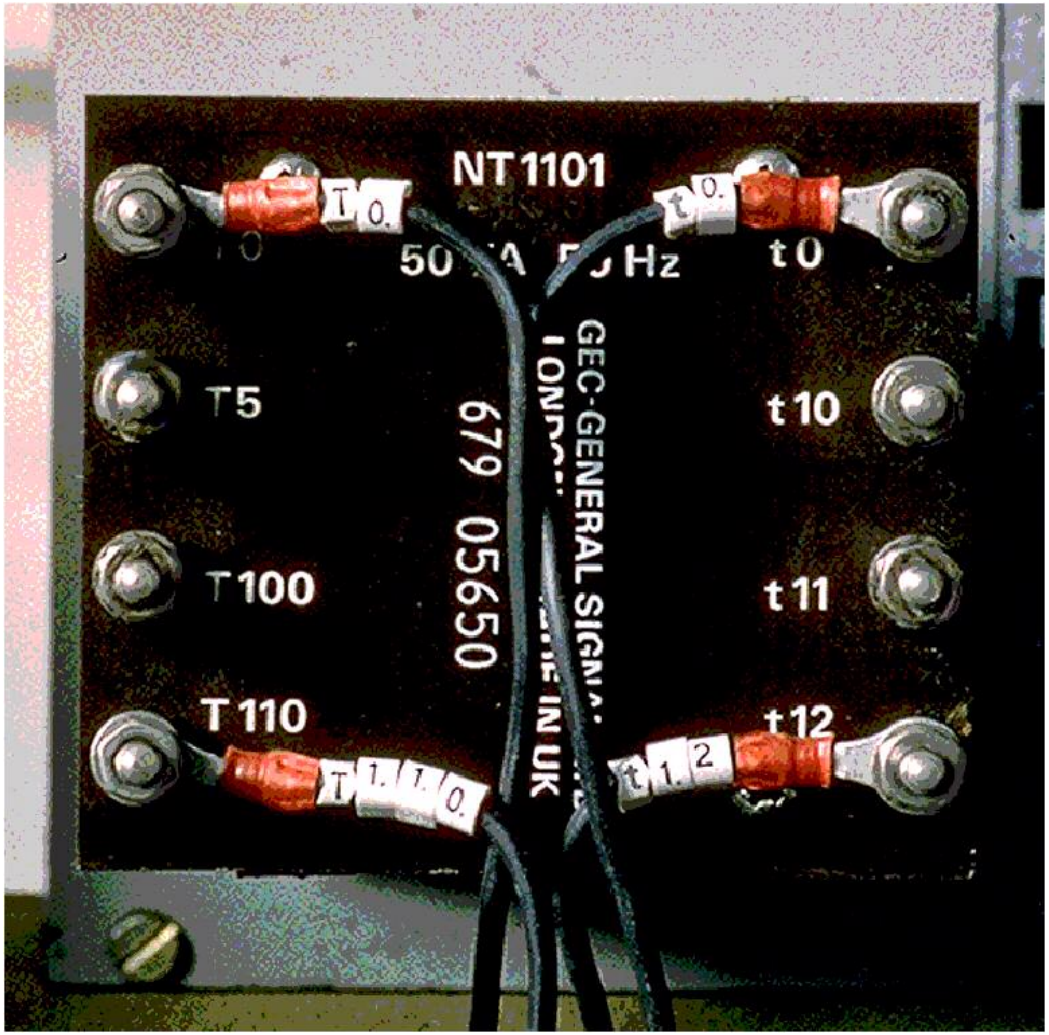


Figure 25

7.5 Self-Assessment Questions

1. Sketch a transformer and identify and label the following parts:

   Primary coil
   Iron core
   Secondary coil
   Input voltage
   Electrical load

2. A transformer has 480 turns in its primary coil and 192 turns in the secondary. If an alternating voltage of 60 volts is applied to the primary coil, what will be the voltage in the secondary coil?

3. What is the function of an isolating transformer?

If you are not totally confident that you have correctly answered any of these questions, re-read the relevant parts of this section to check your answer(s) before going on to the next section.

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8 Transformer Rectifiers

8.1 Need for a Rectifier

A transformer can be used to provide a known value of alternating voltage output but this cannot be used to charge a cell. The charging current must be a direct current and to achieve this a rectifier is needed.

8.2 The Rectifier

A rectifier is a device which allows current to pass in one direction only. When connected to an alternating current supply it will produce a direct current output. A picture of a typical transformer/rectifier is shown in figure 26.

The direct current output from a transformer rectifier is used in the signalling system to

- charge batteries
- supply power to equipment which requires a DC supply

A Transformer/Rectifier

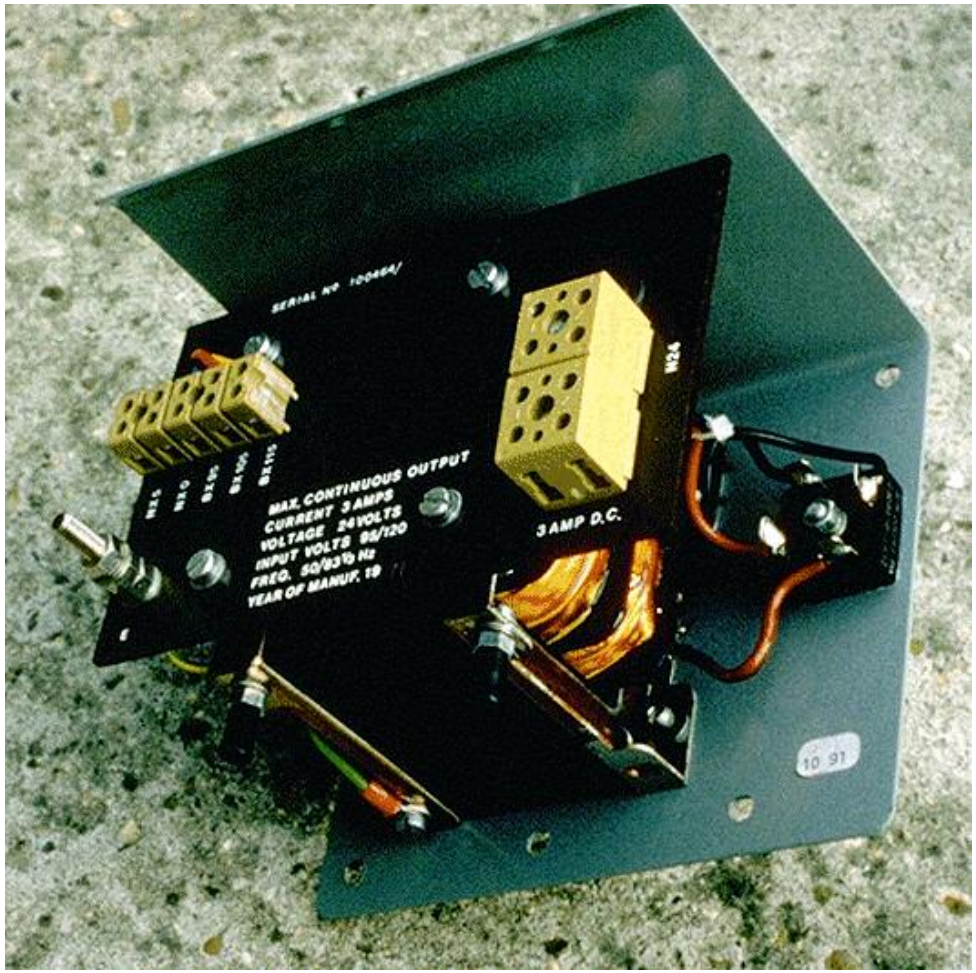


Figure 26

8.3 The Transformer Rectifier

A circuit diagram showing a rectifier connected to a transformer is shown at figure 27. The shape of the voltage waveforms in different parts of the circuit are also shown. The conversion from an alternating to direct current can be clearly seen.

An alternating voltage is applied to the primary coil of the transformer. The turns ratio of the transformer controls the alternating voltage in the secondary coil. The voltage output from the secondary coil is fed to a rectifier circuit which converts it to a direct voltage. This voltage is then fed to the load.

8.4 Assignment 7

You should be able to find examples of transformers and transformer/rectifiers used in your workplace. Identify where they are being used and what they are being used for, and record your findings in the table at figure 28. Speak to your work mates or mentor if you need any help.

8.5 Self-Assessment Questions

1. What is the function of a rectifier?

2. The input to a rectifier is an alternating current. What type of current is the output?

If you are not totally confident you have correctly answered either of these questions, re-read the relevant parts of this section to check your answer(s) before going on to the next section.

Schematic of Transformer/Rectifier Circuit showing Voltage Waveforms

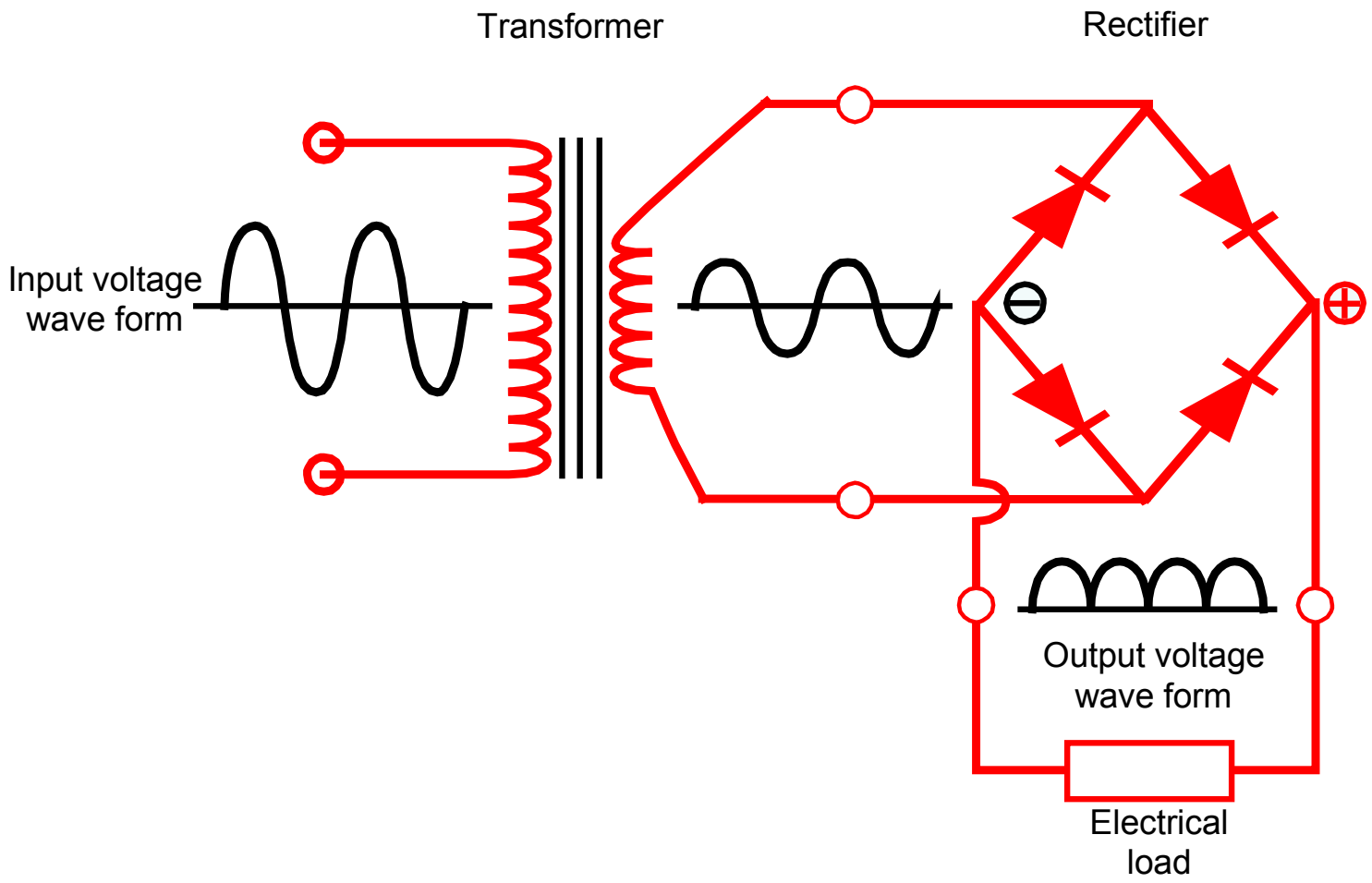


Figure 27

Transformers in the Workplace

Serial	Description of Transformer	Use of Transformer
1		

Transformers/Rectifiers in the Workplace

Serial	Description of Transformer/Rectifier	Use of Transformer/Rectifier
1		

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9 Safety

9.1 Safety and Power Supplies

You must observe appropriate safety precautions at all times when working with power supplies. They are dangerous and must be handled correctly.

The dangers you may encounter will be from:

- Chemicals
- Electricity

Chemical Safety

The chemicals used in cells can cause injury or death from any of the following:

- **Explosion.** The gasses given off by the chemical action within a cell are explosive. Large volumes of gas can be given off during the charging of cells and a continuous flow of fresh air is needed to disperse it safely
- **Poison.** Some of the gasses given off by cells are poisonous and must be well dispersed in the fresh air
- **Electrolyte Burns.** The electrolyte used in wet primary, lead acid and alkaline cells can cause severe burns if it comes into contact with the skin. It can easily burn through your work clothes. Burns to the eyes can cause permanent blindness.

Electrical Safety

All power supplies, whether provided by cells or mains electricity, can be hazardous to your safety. Danger comes from:

- **Electric Shock**. An electric shock can cause heart failure
- **Secondary Injuries**. A person receiving an electric shock can suffer an acute convulsion and be violently thrown away from the source of electricity. This can cause injuries such as cuts, broken bones and concussion
- **Electrical Burns**. These can occur when skin comes into contact with electricity. High voltages can cause severe burning

9.2 Safety Precautions

The safety procedures for handling chemicals are displayed in all areas where chemicals are used. Before you handle any chemical, familiarise yourself with the appropriate safety precautions.

The correct procedures for operating power supply equipment should be read and understood before you use the equipment. This will minimise the risk to you and your work mates.

An example of the safety precautions for batteries is shown in figure 29.

Example of Safety Precautions for Battery Charging

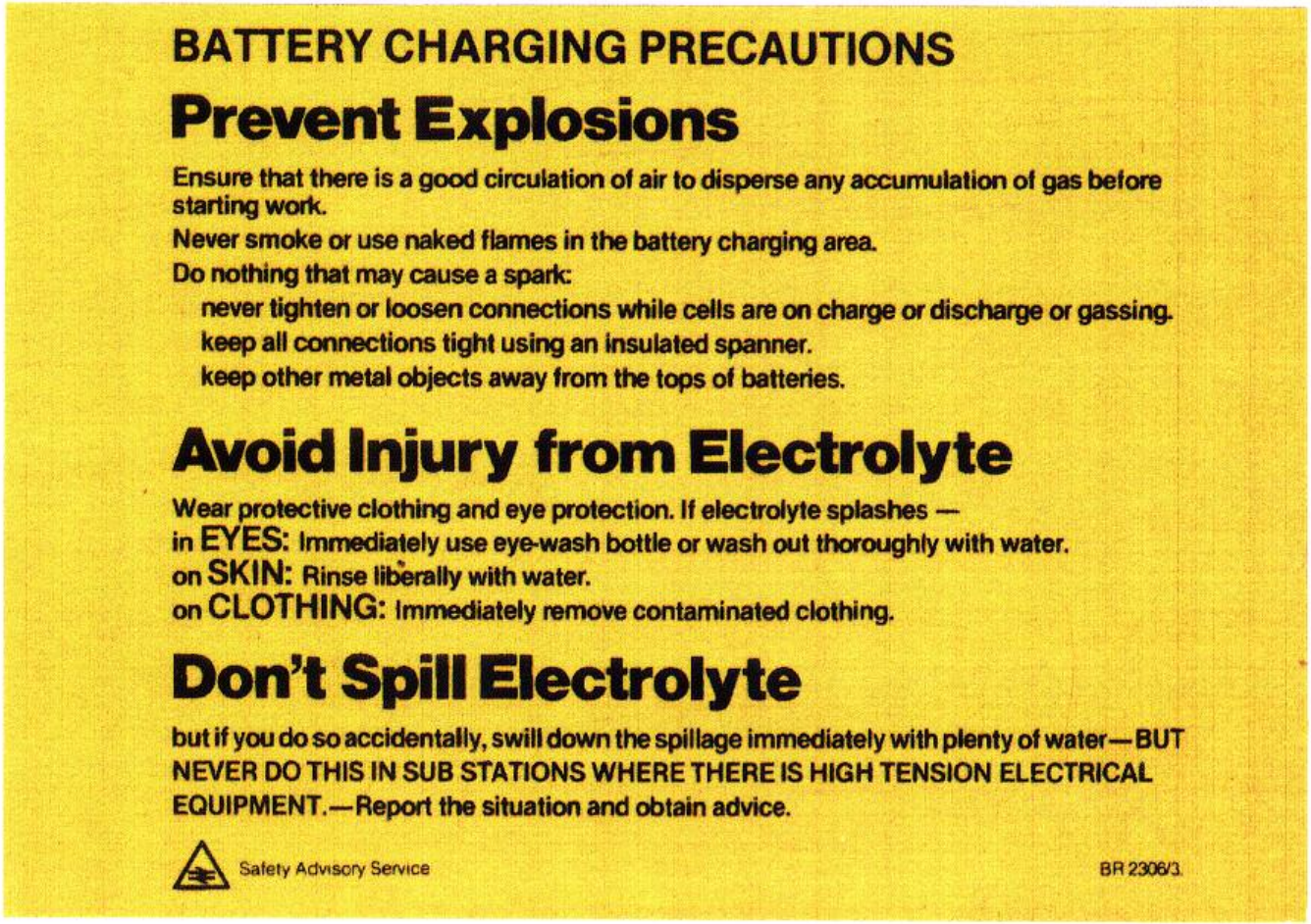


Figure 29

9.3 First Aid

In the event of an accident, you may have to give first aid to a work mate or to yourself. Learn and practise first aid skills.

The following are particularly important when working with power supplies:

- Treatment for injuries caused by electrolyte
- Treatment for electric shock

9.4 Assignment 8

Safety precautions and first aid notices should be displayed in areas where working conditions may be hazardous. Identify hazardous areas in your workplace and read the notices displayed in the areas. If you can not understand any of the precautions or first aid actions, ask your mentor to explain them to you.

9.5 Self-Assessment Questions

1. What are the three main dangers you may be exposed to when working with electrical supplies?
2. What are the three main dangers you must be alert to when working with chemicals?

If you are not totally confident that you have correctly answered any of these questions, re-read the relevant parts of this section to check your answer(s) before going on to the next section.

10 Summary

10.1 Need for Power Supplies

British railways rely heavily on the use of electricity to operate their signalling equipment. In many cases, a direct current supply is required and it is obtained from either of these two sources:

- A **battery**
- A **transformer rectifier**

When an alternating current supply is required it is supplied through a **transformer**.

10.2 Batteries

A **battery** is a self-contained power source which produces electricity by a **chemical reaction** within a **cell**. By connecting an appropriate number of cells together, you can produce a battery of the required output voltage. There are two types of cell, namely:

- **Primary cells**. A discharged primary cell can not be recharged.
- **Secondary cells**. A discharged secondary cell can be recharged.

A primary cell is tested by physically inspecting it and checking its voltage. In addition, the specific gravity of a secondary cell is tested using a hydrometer.

10.3 Transformers

A t**ransformer** is used to supply an alternating voltage to electrical equipment and circuits.

For a given input voltage, the output voltage depends on the number of turns in the **secondary coil** compared to the number of turns in the **primary coil**.

10.4 Transformer Rectifiers

A **transformer rectifier** produces a direct current source of power from an AC supply.

A **rectifier** converts the alternating voltage output of a transformer to a direct voltage.

10.5 Safety

Power supplies, like any other electrical equipment, are safe if you observe the correct **safety precautions**.

You must understand all necessary safety precautions for power supplies, particularly those concerned with electricity and chemicals. You must also be able to carry out **first aid** in the event of an accident.

11 Self-Test

This section contains a test by which you can test your knowledge of the objectives for the Principles of Signalling Power Supplies. The test is of similar duration and standard as that which you will be given in the Module 6 assessment.

The self-test contains questions which require the following types of answer:

True/False. The answer will be either true or false. Circle the correct answer.

Multi choice. You will be offered several answers, only one of which is correct. Circle the identifying letter of the correct answer.

Short Answer. You will need to write down the answer in the space provided.

You should attempt every question. The answers are given in Section 12.

Question 1 A primary cell cannot be recharged. True/false.

Question 2. Three cells, each of 1.4 volts, are connected in series across a load. The voltage supplied to the load is:

Question 3. Two cells, each of which can supply 1.4 volts and 1 amp, are connected in parallel across a load.
The current that can be supplied to the load is:

a. 0.5 amps
b. 1 amp
c. 2 amps

Question 4. The resistance within a cell, which causes a reduction in the terminal voltage, is called:

a. Load resistance
b. Cell resistance
c. Internal resistance

Question 5. The two electrodes of a cell are called:

a. Positive and neutral
b. Negative and positive
c. Negative and neutral

Question 6. The Leclanche Cell is a dry cell. **True/false**

Question 7. One method of testing a primary cell is by physical inspection. State one other method.....

Question 8. The terminal voltage of a cell connected to a constant load is tested with the load shunted by a resistor. **True/false**

Question 9. The lead acid cell is a secondary cell. **True/false**

Question 10. Dilute sulphuric acid is used as the electrolyte of a lead acid cell. **True/false**

Question 11. The specific gravity of distilled water is 0.000. **True/false**

Question 12. The instrument used to measure the specific gravity of electrolyte is a:.....

Question 13. A fully charged lead acid cell has an SG of approximately:

a. 2.220
b. 1.220
c. 0.220

Question 14. An alkaline cell is a primary cell. **True/false**

Question 15. The positive electrode of an alkaline cell is made of nickel hydrate.
The negative electrode is made of:

a. Iron and cadmium
b. Steel and carbon
c. Lead and copper

Question 16. A fully charged alkaline cell has a specific gravity of approximately:.....

Question 17. Write down the three types of charge for a battery:

1. .....

2. .....

3. .....

Question 18. A 72 AH cell is to be charged at the 6-hour rate. The charging current will

be:.....

Question 19. A cell which gives off large amounts of gas when being charged is being charged at:

a. An insufficient charge rate
b. An excessive charge rate
c. The correct charge rate

Question 20. The primary coil of a transformer is connected to the electrical load. **True/false**

Question 21. A transformer has 6500 turns in its primary coil and 1100 turns in its secondary coil. If 650 volts is applied to the primary, the output voltage will be:

a. 110 volts
b. 650 volts
c. 760 volts

Question 22. The function of a rectifier is to change direct voltage to alternating voltage. **True/false**

Question 23. Gasses given off by cells are explosive. **True/false**

Question 24. Electrical burns cause only minor injuries. **True/false**

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12 Answers

Answers to Self-Test Questions

Question 1. True

Question 2. 4.2 volts

Question 3. c

Question 4. c

Question 5. b

Question 6. True

Question 7. Measure terminal voltage

Question 8. False

Question 9. True

Question 10. True

Question 11. False

Question 12. Hydrometer

Question 13. b

Question 14. False

Question 15. a

Question 16. 1.190

Question 17. Initial, trickle, recharge

Question 18. 12 amps

Question 19. b

Question 20. False

Question 21. a

Question 22. False

Question 23. True

Question 24. False

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13 Glossary

a.c. or AC Abbreviation for alternating current.
alternating current A current which alternately flows in one direction and then in the opposite direction.

ampere The unit of electrical current. Usually abbreviated to amp or A.

battery A group of cells connected together.

cell Converts chemical energy into electrical energy.

current Rate of flow of electrical charge. Measured in amperes (A).

d.c. or DC Abbreviation for direct current.

depolariser A chemical included in a cell to prevent polarisation.

direct current Current which flows in one direction only.

electrode A metallic conductor in a cell. A cell has a positive and negative electrode across which the cell's output voltage is produced.

electrolyte A substance which, when dissolved (normally in water), produces a conductingpath in the solvent.

internal resistance The resistance within a cell.

output voltage The voltage across the terminals of a cell when an electrical load connected to it.

parallel circuit A circuit in which all the components have the same voltage across them.

polarisation A chemical effect in a cell which causes the positive electrode to be coated with hydrogen bubbles and so increase the internal resistance of the cell.

primary cell A cell which cannot be recharged.

primary coil The coil of a transformer which is connected to the a.c. supply.

rectifier A circuit which converts alternating voltage or current into direct voltage or current.

secondary cell A cell which can be recharged by passing d.c. through it.

secondary coil The coil of a transformer which is connected to the electrical load.

series circuit A circuit in which all components carry the same current.

shunt resistor A resistor connected in parallel across an electrical load.

transformer A device which transforms or changes voltage levels in an a.c. circuit.

voltage The difference in potential between two points in an electrical circuit. It is measured in